Southwest Tech

Python Programming Foundations

Week 2 - Decision Logic and Repetition

Day 3: "Combining Logic and Repetition"

1

Chose And Repeat

This week, you learned two control-flow tools.

Decisions choose what happens. Loops repeat what happens.

Today, those ideas work together:

- a condition can choose a branch
- a loop can repeat a process
- a condition can also help decide when repetition stops

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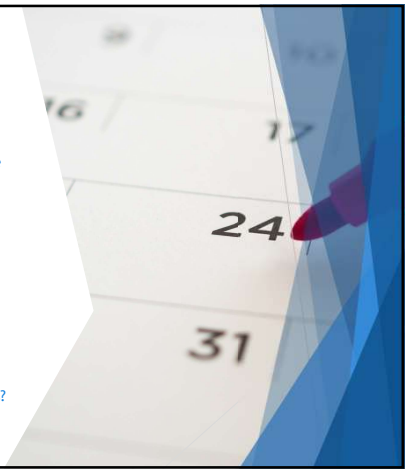
Monday Plus Tuesday

Monday: conditions choose branches.

Tuesday: loops repeat work.

Thursday's job is to keep both ideas explainable:

- What condition is checked?
- What repeats?
- What changes?
- What stops the process?



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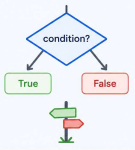
Week Recap

Monday

conditions choose



Tuesday

loops repeat

Thursday: combine carefully

Monday Plus Tuesday

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What Counts as Success Today

A combined program should still be traceable.

PATH: input or value -> condition -> branch or loop -> update -> output

If the path is hard to explain, reduce the program.

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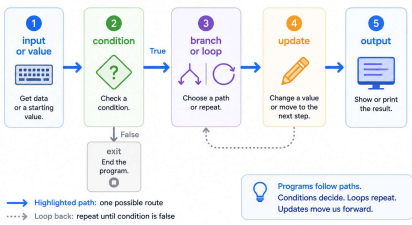
Program Path

A simple path a program follows.



Today's Success Pattern

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What We Will Use Today
conditions inside loops
loops controlled by conditions
simple validation
small menu-style logic
test values and visible output
The goal is controlled combination, not feature count.

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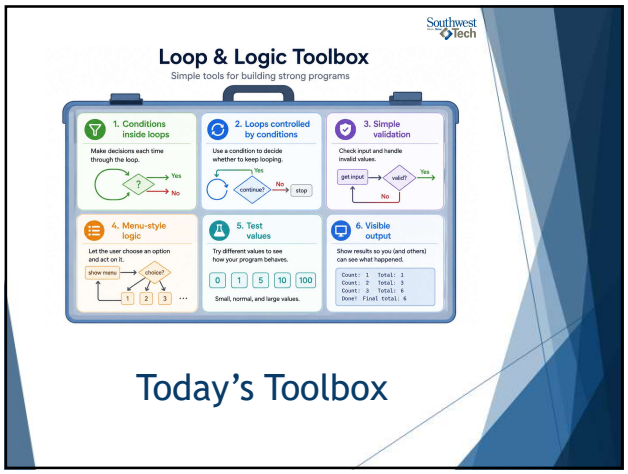
Loop & Logic Toolbox
Simple tools for building strong programs
1. Conditions inside loops
2. Loops controlled by conditions
3. Simple validation
4. Menu-style logic
5. Test values
6. Visible output
Today's Toolbox

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What We Will "Skip" For Now
large interactive applications
complicated nested logic
Avoid adding features until the main path is clear.
Today, one condition plus one repetition pattern is enough for success.

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Parked For Later
These ideas are useful, but they are not today's focus.
SAVE FOR LATER
nested loops
many menu options
multiple features
advanced input handling
Useful ideas, saved until the main path is clear.
"Parked"

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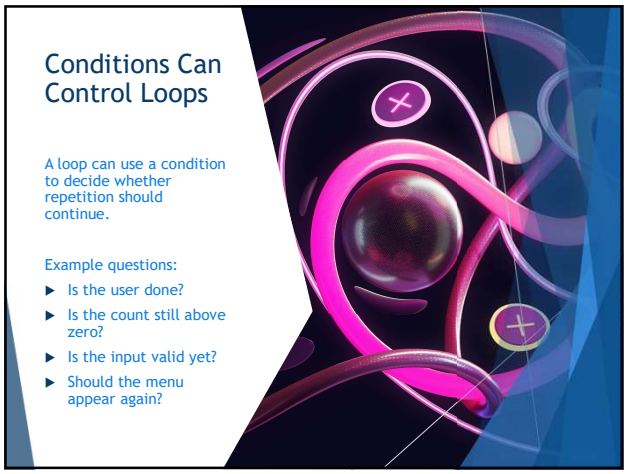
Conditions Can Control Loops
A loop can use a condition to decide whether repetition should continue.
Example questions:
Is the user done?
Is the count still above zero?
Is the input valid yet?
Should the menu appear again?

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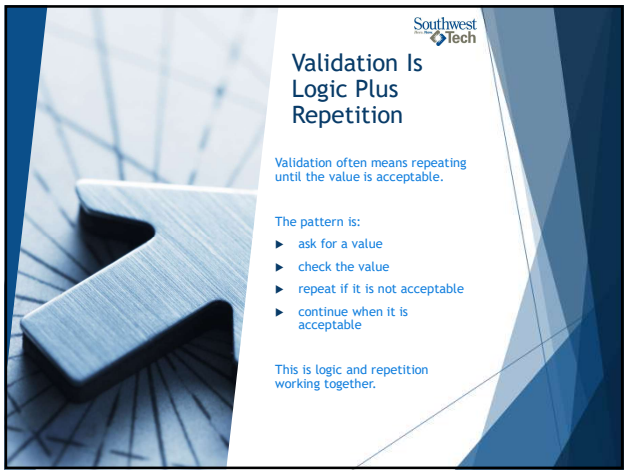
Validation Is Logic Plus Repetition
Validation often means repeating until the value is acceptable.
The pattern is:
ask for a value
check the value
repeat if it is not acceptable
continue when it is acceptable
This is logic and repetition working together.

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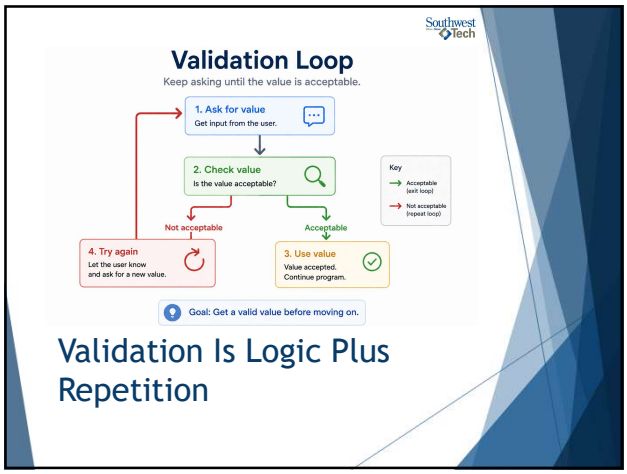

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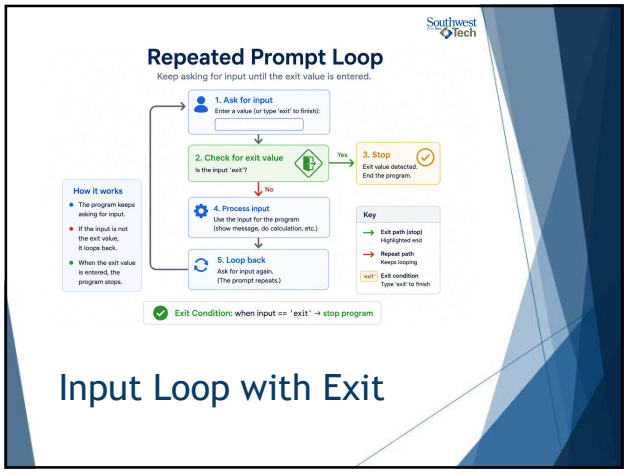

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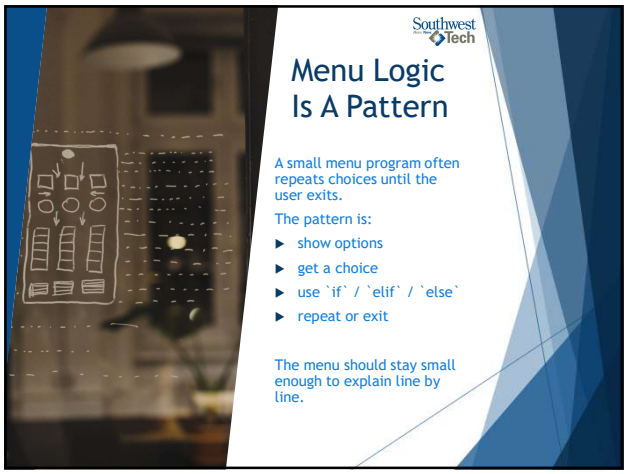

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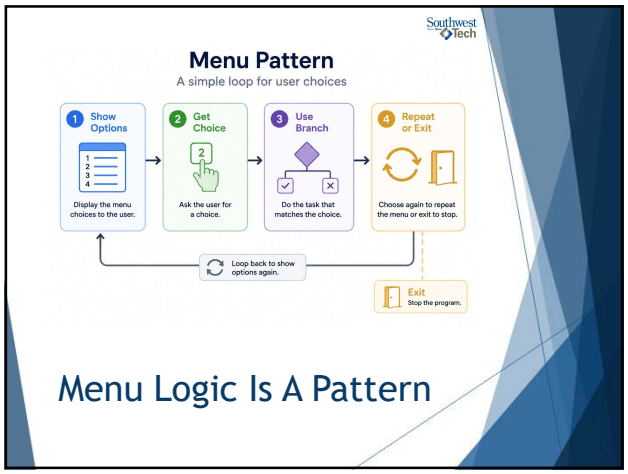

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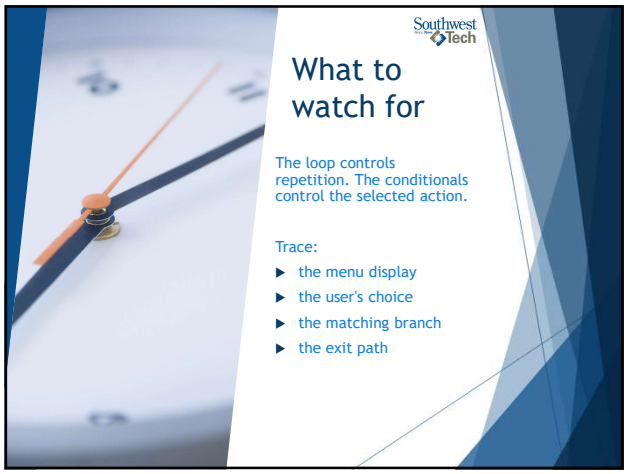

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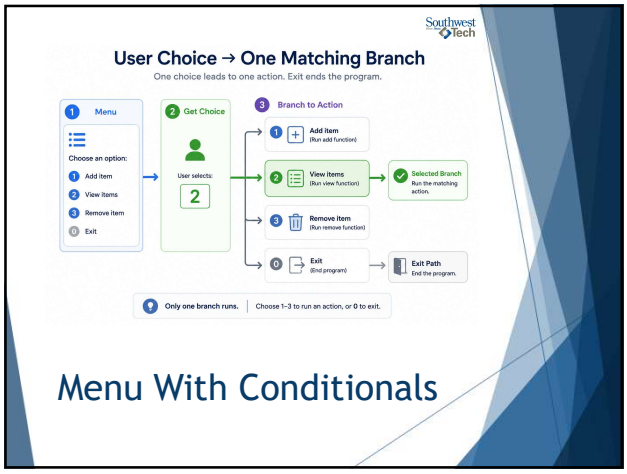

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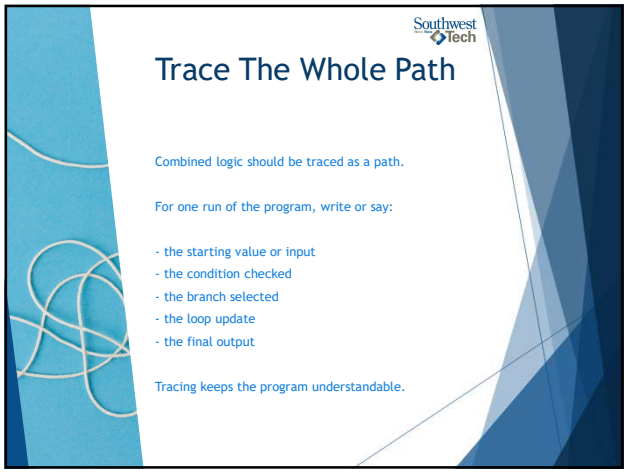

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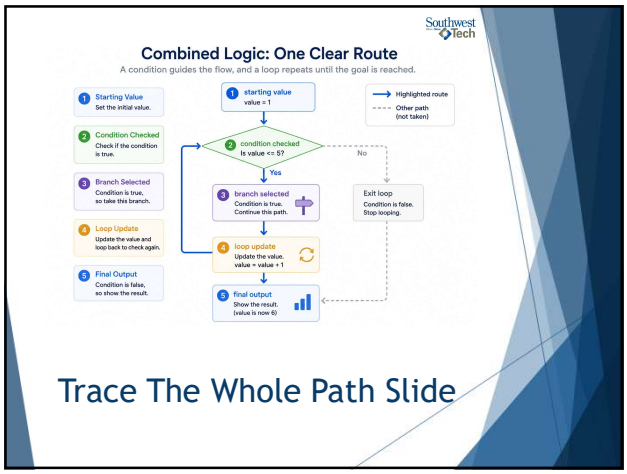

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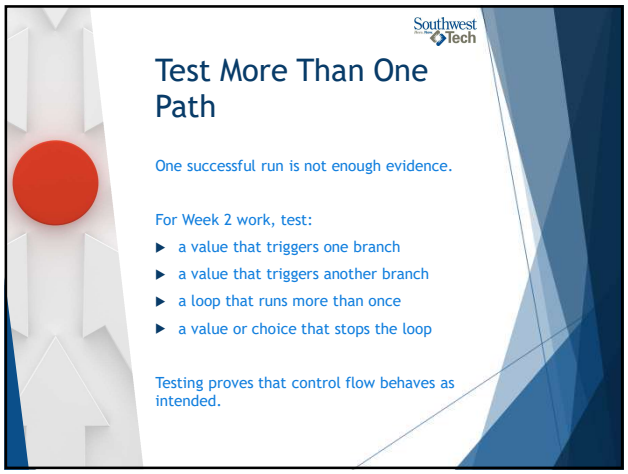

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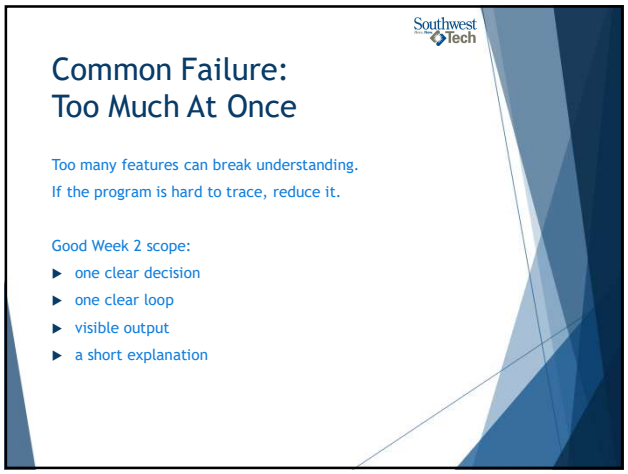

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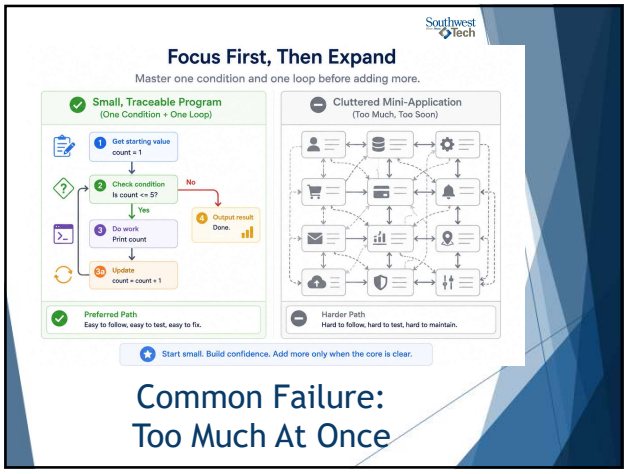

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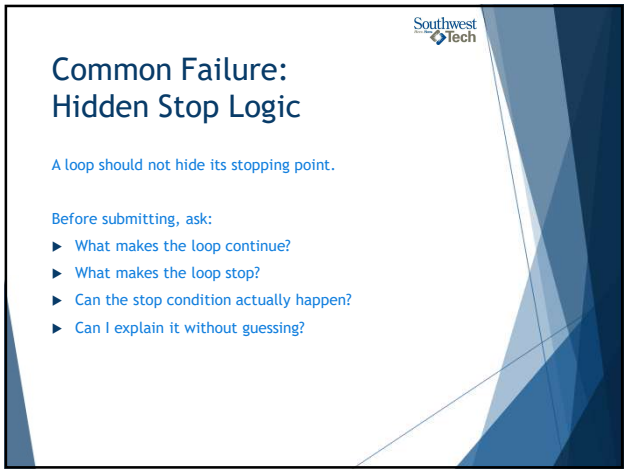

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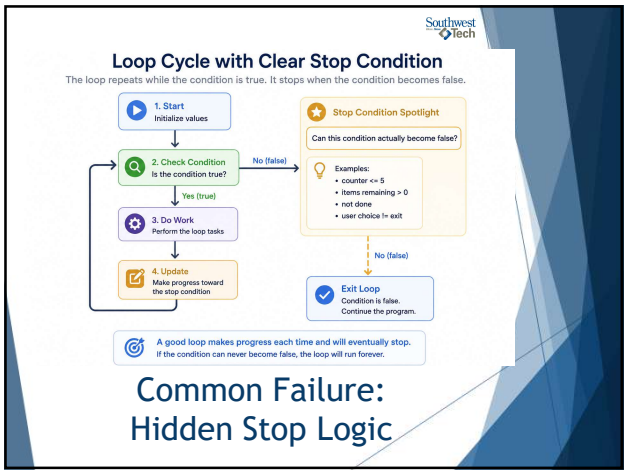

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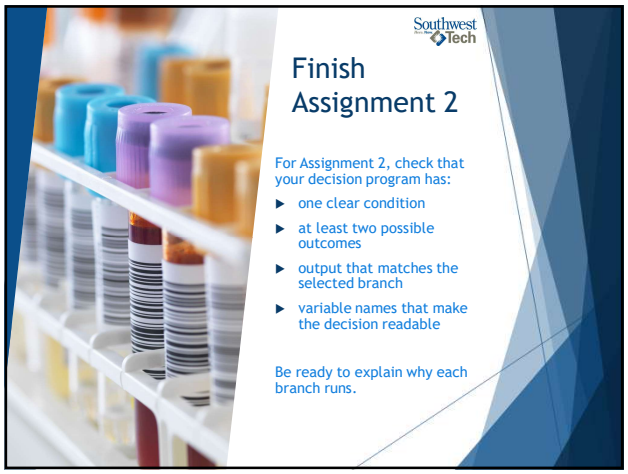

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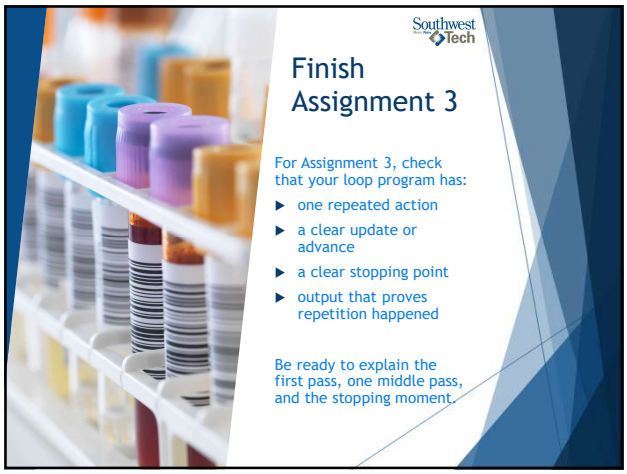

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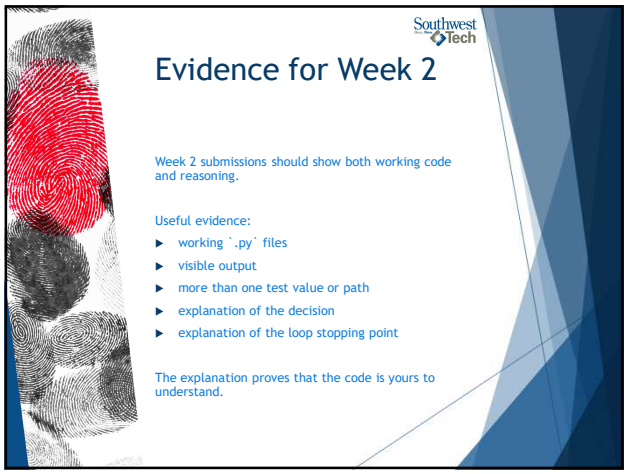

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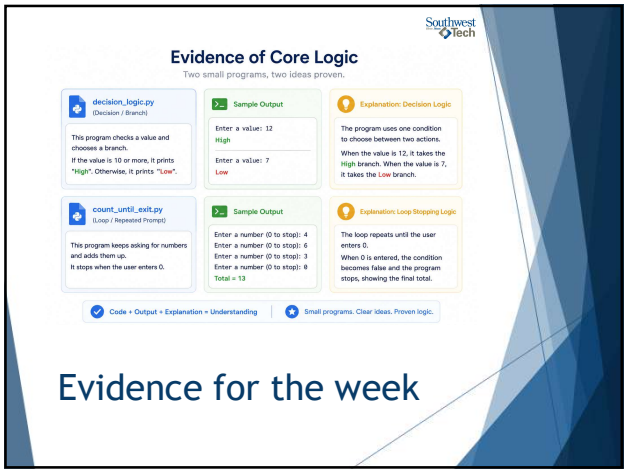

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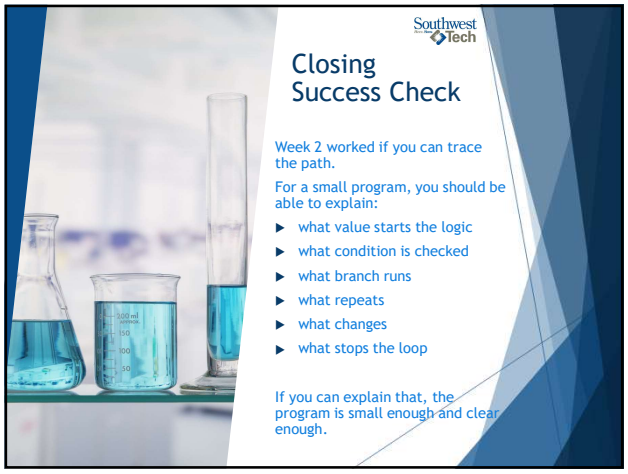

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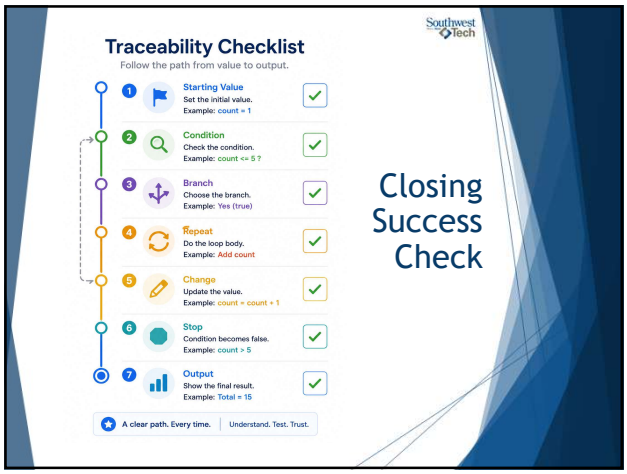

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